

BIP39 24-Word Seed - 100 d6 Roll Worksheet

Use a fair six-sided die. Record every physical roll in order. 99 d6 rolls are slightly under 256 bits; use 100.

Why 100 rolls? $\log_2(6) \times 100 \approx 258.50$ bits; 99 rolls ≈ 255.91 bits. A 24-word BIP39 mnemonic starts from 256 bits of entropy.
Security note: keep completed sheets private. Do not photograph, scan, cloud-sync, email, or type the final seed into an internet-connected device.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Roll quality check

Count of faces (optional sanity check only; do not discard a sequence merely because counts are uneven):

1: _____ 2: _____ 3: _____ 4: _____ 5: _____ 6: _____

Die used / notes: _____

Date (optional): _____ Location (optional): _____

Convert the 100 Rolls to 256 Bits of Entropy

Unbiased rejection sampling preserves a uniform 256-bit result instead of simply truncating or taking an arbitrary modulo.

Step 1 - Convert each die face to a base-6 digit

Map: 1 -> 0 2 -> 1 3 -> 2 4 -> 3 5 -> 4 6 -> 5

Write the resulting 100 base-6 digits, in the same order as your rolls:

001-025: _____

026-050: _____

051-075:

076-100:

Step 2 - Interpret the 100 digits as one base-6 integer X

Equivalent definition: $X = d_1 \times 6^{99} + d_2 \times 6^{98} + \dots + d_{99} \times 6 + d_{100}$, where each d is 0 through 5.

Record X here if you calculate it by hand / offline calculator:

Step 3 - Rejection test (required to avoid bias)

Accept only if:

$$X < 5 \times 2^{256}$$

X < 578960446186580977117854925043439539266349923328202820197287920039565648199680

If X is at or above that threshold, discard all 100 rolls and generate a fresh 100-roll sequence.

A rejection occurs about 11.4% of the time. If accepted, continue.

Step 4 - Derive the 256-bit entropy

For an accepted X , calculate: $E = X \bmod 2^{256}$

Write E as exactly 64 hexadecimal characters (pad on the left with zeroes if needed):

01-16: _____

17-32:

33-48:

49-64: _____

Step 5 - BIP39 checksum

Compute SHA-256(E). For 256-bit entropy, the checksum is the first 8 bits (the first byte) of that SHA-256 digest.

SHA-256(E), first byte (hex): Checksum bits (8):

Step 6 - Make the 264-bit BIP39 stream

Append the 8 checksum bits to the 256 entropy bits. Split the resulting 264 bits into 24 groups of 11 bits.

Each 11-bit group is an index from 0 to 2047. Convert it to 3-digit hex (000-7FF) and look up the word.

The dplusplus table also lists the same word with the high bit set (800-FFF), e.g. 001 or 801 = ability.

Optional offline helper (transcribe; do not run on a networked machine holding real seed material)

```
import hashlib
rolls = '...' # exactly 100 characters, each 1..6
X = 0
for ch in rolls: X = X*6 + (int(ch)-1)
if X >= 5*(1<<256): raise SystemExit('REJECT - reroll all 100 dice')
E = X % (1<<256)
entropy = E.to_bytes(32, 'big')
cs = hashlib.sha256(entropy).digest()[0]
bits = f'{E:0256b}' + f'{cs:08b}'
indices = [int(bits[i:i+11], 2) for i in range(0,264,11)]
hex_codes = [f'{i:03X}' for i in indices]
```

24-Word BIP39 Assembly & Verification Sheet

Do not choose the 24th word independently. Its 11-bit group contains entropy bits plus the 8-bit checksum.

#	11-bit group	Index 0-2047	Hex 000-7FF	BIP39 word
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				

Final verification

- ☐ I used 100 recorded d6 rolls, not 99.
- ☐ The 100-roll base-6 integer passed the rejection test.
- ☐ My entropy is exactly 256 bits / 64 hex characters.
- ☐ I computed the 8-bit checksum from SHA-256(entropy).
- ☐ I split entropy + checksum into exactly twenty-four 11-bit groups.
- ☐ Each index is 0-2047 and matches the intended BIP39 English word.
- ☐ I verified the mnemonic on an offline implementation before receiving funds.

Storage warning:

Anyone who obtains these 24 words can control the wallet. Treat a completed worksheet like cash or a private key.